Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 12.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Simplify with positive exponents only.

$$\left[\frac{(3x^2y^{-3})^2}{(2x^4y)^3}\right]^{-2}$$

A $\frac{9x^{16}y^{18}}{64}$ **B** $\frac{9x^{18}y^{16}}{64}$ **C** $\frac{64x^{16}y^{18}}{81}$ **D** $\frac{64x^{18}y^{16}}{9}$ **E** $\frac{9x^{16}}{64y^{18}}$

2. Lines $f(x)$ and $g(x)$ are parallel. Points $(1,1)$ and $(2,3)$ lie on $f(x)$ while point $(-1,2)$ lies on $g(x)$.

Find $g(x)$.

A $g(x) = \left(\frac{1}{2}\right)x + \frac{1}{2}$ **B** $g(x) = \left(\frac{1}{2}\right)x + \frac{5}{2}$ **C** $g(x) = x + 1$ **D** $g(x) = 2x - 1$ **E** $g(x) = 2x + 4$

3. If both the product and sum of four integers are even, which of the following could be the number of even integers in the group?

I. 0

II. 2

III. 4

A I only **B** II only **C** III only **D** II and III only **E** I, II and III

4. A right-angled triangle has a hypotenuse of $7x - 3$. Its legs are $x + 3$ and $6x$.

Find x.

A 3 **B** 4 **C** 5 **D** 6 **E** 7

5. Basket A contains 4 apples, 6 oranges, and 2 lemons. Basket B contains 5 apples, 4 oranges, and 6 lemons. You are to get two fruits from each basket.

What is the probability of getting 2 oranges from Basket A and then 1 apple and 1 lemon from Basket B?

A $\frac{1}{14}$ **B** $\frac{1}{42}$ **C** $\frac{1}{45}$ **D** $\frac{5}{77}$ **E** $\frac{1}{462}$

6. 910

Which of the following is not a factor of the number shown?

A 2 **B** 3 **C** 5 **D** 7 **E** 13

7. This question is about angles between the clock hands.

What angle is formed by the hands of a clock at $9:20$?

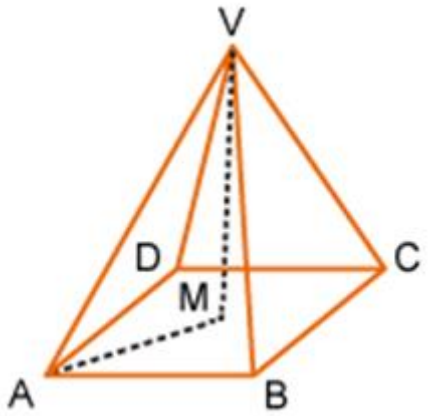
A 160 degrees **B** 150 degrees **C** 145 degrees **D** 130 degrees **E** 30 degrees

8. Look at the diagram below:

The square base $ABCD$ has side lengths of $4\ cm$.

Lengths VA, VB, VC and VD are each $8\ cm$.

M is the mid-point of the square base.



Work out the vertical height in this square-based pyramid. Which is the correct answer?

A $2\sqrt{14}$ **B** $2\sqrt{28}$ **C** $\sqrt{14}$ **D** $\sqrt{58}$ **E** $2\sqrt{56}$

9. A mathematics teacher attempts to solve a difficult calculation without using a calculator. He puts the calculation on the board for his pupils to try.

The calculation is $3424 \times 3427 - 3425 \times 3425$. What is the correct answer?

A 1 **B** 2 **C** 3423 **D** 3044 **E** 11727198

10. When a number a is added to 2137, the result is divisible by both 11 and 23

What is the smallest number that a could possibly be?

A 139 **B** 12 **C** 59 **D** 8 **E** 140

11. When a number d is added to 6789, the number generated is divisible by both 7 and 13

What is the smallest number that d could possibly be?

A 21 **B** 8 **C** 32 **D** 36 **E** 42

12. A teacher attempts a difficult calculation. He puts the question on the board so his students can attempt it at the same time as him.

The calculation is $189 + 190 \times 191 - 188 \times 190$. What is the solution to the teacher's problem?

A 778 **B** 756 **C** 759 **D** 782 **E** 734

13. $b + 12 = \sqrt{(\sqrt{324} + \sqrt{324})}/(\frac{4}{\sqrt{400}})$

Solve the equation to find b?

A 15 **B** 16 **C** 17 **D** 18 **E** 19

14. Simplify $4 - \frac{17(x^2-9)}{5x^2+17x+6}$

A 24 **B** $\frac{3x-43}{5x+2}$ **C** $\frac{3x+59}{5x+2}$ **D** 5 **E** $\frac{3x^2+68x+177}{5x^2+17x+6}$

15. Fred draws an isosceles triangle. The length of the shortest side of the triangle is $8cm$.

The area of the triangle is $32cm^2$. What is the perimeter of the triangle?

A $\sqrt{80}$ **B** $8\sqrt{5} + 8$ **C** $8 + 4\sqrt{5}$ **D** 6 **E** 12

16. Two cars are having a drag race. Both of the cars - Car A and Car B start at the same time and complete the $100\ m$ drag strip at different times. Car A accelerates at a constant speed of $3.2\ m/s$ and finishes the 100 in front of Car B.

Car B accelerates at a constant speed of $3m/s$. How many meters behind car A, was car B when it crossed the finish line?

A $2.5m$ **B** $3.34m$ **C** $5m$ **D** $7.5m$ **E** $10m$

17. The price of a share of company A's stock fell by 20 percent two weeks ago and by another 25 percent last week to its current price. By what percent of the current price does the share need to rise in order to return to its original price?

A 45% **B** 55% **C** $66\frac{2}{3}\%$ **D** 75% **E** 82%

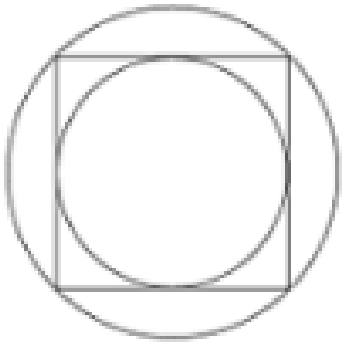
18. $\diamond\ x$ is defined as the largest even number that is less than the negative square root of x. What is the value of $\diamond\ 81$?

A -82 **B** -80 **C** -10 **D** -8 **E** 8

19. If $a > 1$, the ratio of $2a + 6$ to $a^2 + 2a - 3$ is

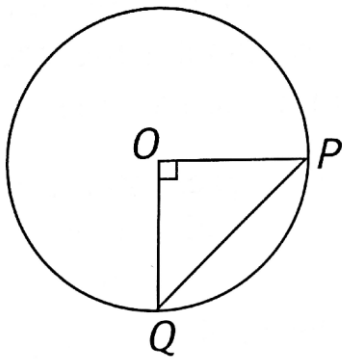
A $2a$ **B** $a + 3$ **C** $\frac{2}{a-1}$ **D** $\frac{2a}{3(3-a)}$ **E** $\frac{a-1}{2}$

20. In the figure, the smaller circle is inscribed in the square and the square is inscribed in the larger circle. If the length of each side of the square is s, what is the ratio of the area of the larger circle to the area of the smaller circle?



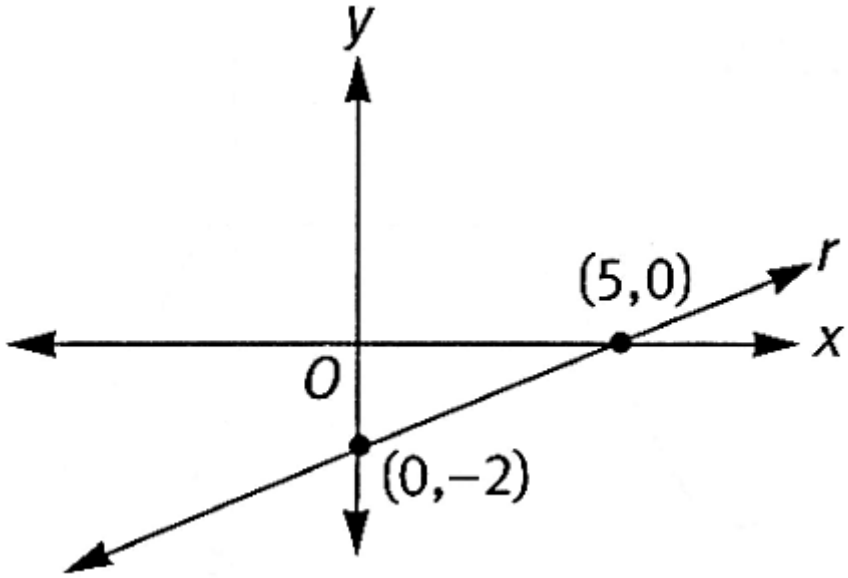
A $2\sqrt{2}:1$ **B** $2:1$ **C** $\sqrt{2}:1$ **D** $2s:1$ **E** $s\sqrt{2}:1$

21. If the length of chord $PQ = 4\sqrt{2}$, what is the circumference of the circle with center O?



A 4 **B** 8 **C** 4π **D** 8π **E** $8\pi\sqrt{2}$

22. Line r is s straight line as shown. Which of the following points lies on line r?



A $(6,6)$ **B** $(7,3)$ **C** $(8,2)$ **D** $(9,3)$ **E** $(10,2)$

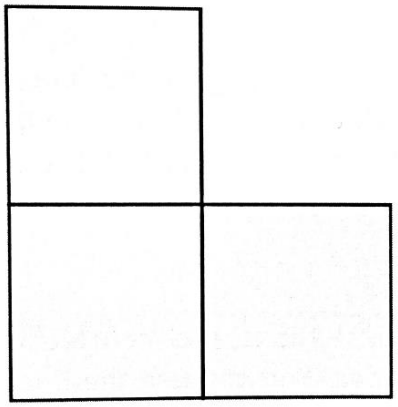
23. If $r = 3s$, $s = 5t$, $t = 2u$, and $u \neq 0$, what is the value of $\frac{rst}{u^3}$?

A 30 **B** 60 **C** 150 **D** 300 **E** 600

24. At Central Park Zoo, the ratio of sea lions to penguins is $4:11$. If there are 84 more penguins than sea lions, how many sea lions are there?

A 24 **B** 36 **C** 48 **D** 72 **E** 121

25. The figure is made up of 3 squares. If the perimeter of the figure is 40 units, what is the area of the figure in square units?

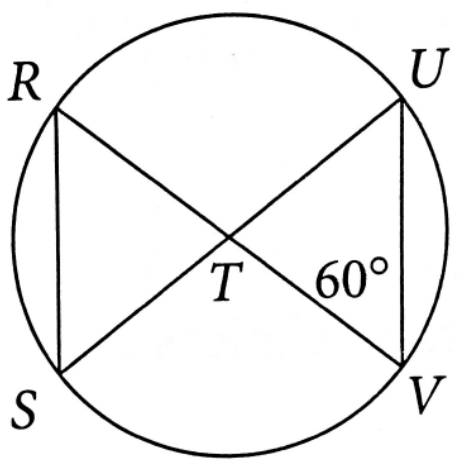


A 50 **B** 75 **C** 120 **D** 150 **E** 200

26. Set T consists of five integers: the first five odd prime numbers when counting upward from zero. This gives set T a standard deviation of approximately 3.71. Which of the following values, if added to the set T, would increase the standard deviation of set T?

A 11 **B** 9 **C** 7.8 **D** 4.15 **E** 3.7

27. The circle shown has center T. The measure of angle TVU is $60°$. If the circle has a radius 3, what is the length of segment RS?



A 2 **B** $2\sqrt{2}$ **C** 3 **D** $3\sqrt{3}$ **E** $6\sqrt{2}$

28. What is the probability of rolling a total of 7 with a single roll of two fair six-sided dice, each with the distinct numbers $1-6$ on each side?

A $\frac{1}{12}$ **B** $\frac{1}{6}$ **C** $\frac{2}{7}$ **D** $\frac{1}{3}$ **E** $\frac{1}{2}$

29. If it takes three days for 10 workers to finish building one house, how many days will it take 15 workers to finish four houses?

A 15 **B** 12 **C** 8 **D** 5 **E** 16

30. What is the average of $2x+3$, $5x-4$, $6x-6$, and $3x-1$?

A $2x+4$ **B** $3x-2$ **C** $3x+2$ **D** $4x-2$ **E** $4x+2$

Answer key:

№	Correct Answer	Your answer
1	C	
2	E	
3	D	
4	B	
5	D	
6	B	
7	A	
8	A	
9	C	
10	E	
11	D	
12	C	
13	D	
14	C	
15	B	
16	A	
17	C	
18	C	
19	C	
20	B	
21	D	
22	E	
23	E	
24	C	
25	B	
26	E	
27	C	
28	B	
29	C	
30	D	
		Total: _____